Lecture 20

Real Forms and Noncompact Structure

Objectives. *A connected real semisimple group with finite centre has a product decomposition into a compact subgroup, an abelian subgroup and a nilpotent subgroup. The orbit theorem of Lecture 19 concerns nilpotent groups and does not apply to such a group, so we start from its structure. This lecture compares compact and split real forms and states the Cartan and Iwasawa decompositions. It proves both decompositions for* $\mathrm{SL}(2,\mathbb{R})$*, where they give coordinates on the upper half-plane.*

By the end of this lecture you will be able to:

- Distinguish the compact and split examples of real forms using their Killing forms and Cartan involutions.
- State the Cartan and Iwasawa decompositions with all their hypotheses.
- Compute $g = kan$ and $g = k\exp X$ for an explicit matrix in $\mathrm{SL}(2,\mathbb{R})$.
- Read both decompositions as coordinates on the upper half-plane.

20.1 Real forms and Cartan involutions

Throughout this section $\mathfrak{g}_{\mathbb{C}}$ is a complex semisimple Lie algebra with Killing form

$$\kappa(X,Y) = \operatorname{tr}(\operatorname{ad}X\ \operatorname{ad}Y).$$

We cite Knapp, *Lie Groups Beyond an Introduction*, digital second edition 2023, as Knapp Beyond.

Definition 20.1. A *real form* of $\mathfrak{g}_{\mathbb{C}}$ is a real Lie subalgebra $\mathfrak{g} \subset \mathfrak{g}_{\mathbb{C}}$ with $\mathfrak{g}_{\mathbb{C}} = \mathfrak{g} \oplus i\mathfrak{g}$ as real vector spaces.

A real basis of $\mathfrak{g}$ is then a complex basis of $\mathfrak{g}_{\mathbb{C}}$. For $X \in \mathfrak{g}$, the matrix of $\operatorname{ad}X$ in this basis is real. It is the same over $\mathbb{R}$ and over $\mathbb{C}$. Hence the Killing form of $\mathfrak{g}$ is the restriction $\kappa|_{\mathfrak{g}}$. It is real and nondegenerate. Its signature is unchanged by real isomorphisms.

Definition 20.2. A real form $\mathfrak{g}$ is *compact* if $\kappa|_{\mathfrak{g}}$ is negative definite. It is *split* if it contains an abelian subalgebra $\mathfrak{h}_0$ with the following two properties. Every $\operatorname{ad}H$ with $H \in \mathfrak{h}_0$ is diagonalisable over $\mathbb{R}$. The sum $\mathfrak{h}_0 \oplus i\mathfrak{h}_0$ is a Cartan subalgebra of $\mathfrak{g}_{\mathbb{C}}$.

Every complex semisimple Lie algebra has both types. The Lie Algebras course does not treat real forms, so we cite this fact from Knapp Beyond, Section VI.1 (Theorem 6.6, Corollary 6.10 and Theorem 6.11).

The construction starts from a *Chevalley basis*. This basis consists of the coroots H_α and of root vectors E_α with $[E_\alpha, E_{-\alpha}] = H_\alpha$. Its structure constants are real, and the linear map given by $E_\alpha \mapsto -E_{-\alpha}$ and $H_\alpha \mapsto -H_\alpha$ is an automorphism. Knapp scales these vectors by real factors, which does not change the real spans below. The real span of the Chevalley basis is a split form, with $\mathfrak{h}_0$ the real span of the H_α. The elements iH_α, $E_\alpha - E_{-\alpha}$ and $i(E_\alpha + E_{-\alpha})$ span a compact form. For $\mathfrak{sl}(2,\mathbb{C})$, with $E_\alpha = E$, $E_{-\alpha} = F$ and $H_\alpha = H$, these elements are the matrices A, B and C of the next example.

Example 20.3 proves both claims for $\mathfrak{sl}(2,\mathbb{C})$, and Proposition 20.5 proves them for $\mathfrak{sl}(n,\mathbb{C})$. The general case stays cited.

The name “compact” refers to compact groups. By Lecture 6, the Lie algebra of a compact group has an Ad-invariant inner product, so each $\operatorname{ad} X$ is skew and

$$\kappa(X,X) = \operatorname{tr}\left((\operatorname{ad} X)^2\right) \le 0.$$

So a real form whose Killing form takes a positive value is not the Lie algebra of a compact group.

Example 20.3 (The two forms of $\mathfrak{sl}(2,\mathbb{C})$)**.** On $\mathfrak{sl}(2,\mathbb{C})$ one has $\kappa(X,Y) = 4\operatorname{tr}(XY)$. Take

$$H = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad E = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad F = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}.$$

Then $H^2 = I$, $(E+F)^2 = I$ and $(E-F)^2 = -I$. The products

$$H(E+F) = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad H(E-F) = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad (E-F)(E+F) = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

are traceless. In the basis $(H, E+F, E-F)$ of $\mathfrak{sl}(2,\mathbb{R})$ the Gram matrix of κ is therefore $\operatorname{diag}(8,8,-8)$. The signature is $(2,1)$. The real form $\mathfrak{sl}(2,\mathbb{R})$ is split with $\mathfrak{h}_0 = \mathbb{R}H$, because $\operatorname{ad} H$ has eigenvalues $2, 0, -2$ on E, H, F.

The course basis of $\mathfrak{su}(2)$ is

$$A = iH, \qquad B = E - F, \qquad C = i(E+F).$$

Thus $\kappa(A,A) = i^2 \cdot 8 = -8$, $\kappa(C,C) = -8$ and $\kappa(B,B) = -8$. The cross terms are multiples of the vanishing ones above. The Gram matrix is $-8I$, so $\mathfrak{su}(2)$ is compact. The signatures $(2,1)$ and $(0,3)$ differ, so the two real forms are not isomorphic.

The two real forms in the example share the line $\mathbb{R}B$. The split form adds the plane $\operatorname{span}(H, E+F)$. The compact form adds the plane $i\operatorname{span}(H, E+F)$, spanned by A and C. Thus

$$\mathfrak{sl}(2,\mathbb{R}) = \mathbb{R}B \oplus \operatorname{span}(H, E+F), \qquad \mathfrak{su}(2) = \mathbb{R}B \oplus i\operatorname{span}(H, E+F).$$

A Cartan involution makes this relation precise. Let $\mathfrak{g}$ now be any real semisimple Lie algebra. Write $\kappa(X,Y) = \operatorname{tr}(\operatorname{ad} X \operatorname{ad} Y)$ for the Killing form of $\mathfrak{g}$ itself. For a real form of $\mathfrak{g}_{\mathbb{C}}$, this is the restriction $\kappa|_{\mathfrak{g}}$ described above.

Definition 20.4. A *Cartan involution* of $\mathfrak{g}$ is a Lie algebra automorphism θ with $\theta^2 = \mathrm{id}$ such that

$$\kappa_\theta(X,Y) = -\kappa(X,\theta Y)$$

is positive definite on $\mathfrak{g}$.

The form κ_θ is symmetric. Since θ is an automorphism, $[\theta X, Z] = \theta[X, \theta^{-1}Z]$ for all $Z \in \mathfrak{g}$. So $\operatorname{ad}(\theta X) = \theta \operatorname{ad} X\, \theta^{-1}$. Hence $\kappa(\theta X, \theta Y) = \operatorname{tr}(\theta \operatorname{ad} X \operatorname{ad} Y\, \theta^{-1}) = \kappa(X, Y)$. Using $\theta^2 = \mathrm{id}$ and the symmetry of κ, we get

$$\kappa(X, \theta Y) = \kappa(\theta X, \theta^2 Y) = \kappa(\theta X, Y) = \kappa(Y, \theta X).$$

Thus $\kappa_\theta(X, Y) = \kappa_\theta(Y, X)$. So κ_θ is an inner product on $\mathfrak{g}$, as the spectral theorem in § 20.2 requires.

Write $\mathfrak{k}$ and $\mathfrak{p}$ for the $(+1)$- and (-1)-eigenspaces of θ. Since $\theta^2 = \mathrm{id}$, we have $\mathfrak{g} = \mathfrak{k} \oplus \mathfrak{p}$. Because θ is an automorphism,

$$[\mathfrak{k}, \mathfrak{k}] \subset \mathfrak{k}, \qquad [\mathfrak{k}, \mathfrak{p}] \subset \mathfrak{p}, \qquad [\mathfrak{p}, \mathfrak{p}] \subset \mathfrak{k}.$$

For instance, $X, Y \in \mathfrak{p}$ give $\theta[X, Y] = [-X, -Y] = [X, Y]$. We cite from Knapp Beyond, Section VI.2, that every real semisimple Lie algebra has a Cartan involution, unique up to inner automorphism.

Proposition 20.5. *Let $\mathfrak{g} = \mathfrak{sl}(n, \mathbb{R})$ and $\theta(X) = -X^{T}$.*

(a) θ is a Cartan involution. Here $\mathfrak{k} = \mathfrak{so}(n)$, and $\mathfrak{p}$ consists of the real symmetric traceless matrices.

(b) $\mathfrak{k} \oplus i\mathfrak{p} = \mathfrak{su}(n)$, and $\mathfrak{su}(n)$ is a compact real form of $\mathfrak{sl}(n, \mathbb{C})$.

(c) The traceless diagonal matrices form a maximal abelian subspace $\mathfrak{a} \subset \mathfrak{p}$. Each $\operatorname{ad} H$, $H \in \mathfrak{a}$, is diagonalisable over $\mathbb{R}$. Hence $\mathfrak{sl}(n, \mathbb{R})$ is split.

Proof. (a) Clearly $\theta^2 = \mathrm{id}$. It is an automorphism because

$$[\theta X, \theta Y] = X^{T}Y^{T} - Y^{T}X^{T} = (YX - XY)^{T} = \theta[X, Y].$$

On $\mathfrak{sl}(n, \mathbb{C})$ we import $\kappa(X, Y) = 2n \operatorname{tr}(XY)$. Then

$$\kappa_\theta(X, X) = 2n \operatorname{tr}(XX^{T}) = 2n \sum_{i,j} X_{ij}^2 > 0 \qquad (X \neq 0).$$

The eigenspace descriptions are immediate.

(b) Write $Z \in \mathfrak{gl}(n, \mathbb{C})$ as $Z = X + iY$ with X, Y real. Then $Z^{*} = X^{T} - iY^{T}$. So $Z^{*} = -Z$ exactly when $X^{T} = -X$ and $Y^{T} = Y$. Also $\operatorname{tr} Z = 0$ exactly when $\operatorname{tr} Y = 0$, since $\operatorname{tr} X = 0$ already. Hence $\mathfrak{su}(n) = \mathfrak{k} \oplus i\mathfrak{p}$. It is a real form because

$$\mathfrak{sl}(n, \mathbb{C}) = (\mathfrak{k} \oplus \mathfrak{p}) \oplus i(\mathfrak{k} \oplus \mathfrak{p}) = (\mathfrak{k} \oplus i\mathfrak{p}) \oplus i(\mathfrak{k} \oplus i\mathfrak{p}).$$

For nonzero $Z \in \mathfrak{su}(n)$, we get $\kappa(Z, Z) = 2n \operatorname{tr}(Z^2) = -2n \operatorname{tr}(Z^{*}Z) < 0$.

(c) Let $H = \operatorname{diag}(h_1, \dots, h_n)$ and let E_{ij} be the matrix units. Then

$$[H, E_{ij}] = (h_i - h_j) E_{ij}.$$

So $\operatorname{ad} H$ is diagonal, with real eigenvalues, in the basis of all E_{ij} with $i \neq j$ together with a basis of $\mathfrak{a}$. For maximality, choose $H \in \mathfrak{a}$ with distinct entries. If $Y \in \mathfrak{p}$ commutes with H, then $(h_i - h_j) Y_{ij} = 0$, so Y is diagonal. Finally, $\mathfrak{a} \oplus i\mathfrak{a}$ is the diagonal Cartan subalgebra of $\mathfrak{sl}(n, \mathbb{C})$. ∎

The recipe $\mathfrak{k} \oplus i\mathfrak{p}$ from the compact-form statement in Proposition 20.5 works in general. If θ is a Cartan involution of $\mathfrak{g}$, then $\mathfrak{k} \oplus i\mathfrak{p}$ is a compact real form of the complexification $\mathfrak{g}_{\mathbb{C}}$ of $\mathfrak{g}$, the complex Lie algebra in which $\mathfrak{g}$ is a real form. This general fact is cited from Knapp Beyond, Sections VI.1–VI.2. For $\mathfrak{g} = \mathfrak{sl}(2,\mathbb{R})$ this recipe returns the decomposition of $\mathfrak{su}(2)$ displayed above.

20.2 Restricted roots and the global decompositions

Fix a real semisimple Lie algebra $\mathfrak{g}$ with Cartan involution θ, and write $\mathfrak{g} = \mathfrak{k} \oplus \mathfrak{p}$.

Lemma 20.6. *For $X \in \mathfrak{p}$, the operator $\operatorname{ad} X$ is symmetric for κ_θ.*

Proof. Since θ is an automorphism and $\theta X = -X$, we have $[X, \theta Z] = -\theta[X,Z]$. Invariance of κ gives $\kappa([X,Y],W) = -\kappa(Y,[X,W])$. Hence

$$\kappa_\theta([X,Y],Z) = \kappa(Y,[X,\theta Z]) = -\kappa(Y,\theta[X,Z]) = \kappa_\theta(Y,[X,Z]).$$

∎

Choose a maximal abelian subspace $\mathfrak{a} \subset \mathfrak{p}$. The operators $\operatorname{ad} H$, $H \in \mathfrak{a}$, commute and are κ_θ-symmetric. By the spectral theorem they are simultaneously diagonalisable over $\mathbb{R}$.

Definition 20.7. For $\lambda \in \mathfrak{a}^*$ put

$$\mathfrak{g}_\lambda = \{X \in \mathfrak{g} : [H,X] = \lambda(H)X \text{ for all } H \in \mathfrak{a}\}.$$

A *restricted root* is a nonzero λ with $\mathfrak{g}_\lambda \neq 0$. The *real rank* of $\mathfrak{g}$ is $\dim \mathfrak{a}$.

The name refers to the roots of the complexification $\mathfrak{g}_{\mathbb{C}}$. For a Cartan subalgebra of $\mathfrak{g}_{\mathbb{C}}$ containing $\mathfrak{a}$, each restricted root is the restriction to $\mathfrak{a}$ of a root of $\mathfrak{g}_{\mathbb{C}}$. For example, § 20.3 takes $\mathfrak{g} = \mathfrak{sl}(2,\mathbb{R})$ and $\mathfrak{a} = \mathbb{R}H$, and finds the restricted roots $\pm\lambda_1$ with $\lambda_1(H) = 2$.

Fix $H_0 \in \mathfrak{a}$ with $\lambda(H_0) \neq 0$ for the finitely many restricted roots. Such an H_0 exists, because $\mathfrak{a}$ is not a union of finitely many hyperplanes $\ker\lambda$. Call λ positive if $\lambda(H_0) > 0$, and set

$$\mathfrak{n} = \bigoplus_{\lambda(H_0)>0} \mathfrak{g}_\lambda.$$

The Jacobi identity gives $[\mathfrak{g}_\lambda, \mathfrak{g}_\mu] \subset \mathfrak{g}_{\lambda+\mu}$. Hence $\mathfrak{n}$ is a subalgebra. Each $\operatorname{ad} X$ with $X \in \mathfrak{n}$ is nilpotent, because it strictly raises the value at H_0.

For $\mathfrak{su}(n)$ the identity is a Cartan involution, since $\kappa_{\mathrm{id}} = -\kappa$ is positive definite. The two forms of $\mathfrak{sl}(n,\mathbb{C})$ compare as follows.

Real form	θ	$\mathfrak{k}$	$\mathfrak{p}$	real rank
$\mathfrak{su}(n)$, compact	id	$\mathfrak{su}(n)$	0	0
$\mathfrak{sl}(n,\mathbb{R})$, split	$X \mapsto -X^{T}$	$\mathfrak{so}(n)$	symmetric traceless	$n-1$

For $\mathfrak{sl}(n,\mathbb{R})$, with $H_0 = \operatorname{diag}(n-1, n-3, \dots, 1-n)$, the algebra $\mathfrak{n}$ is the strictly upper triangular matrices (Exercise 20.2).

The real rank in the table does not depend on the choices of θ and $\mathfrak{a}$. We cite this from Knapp Beyond, Chapter VI. First fix θ. Any two maximal abelian subspaces of $\mathfrak{p}$ are then conjugate by a finite product of automorphisms $e^{\operatorname{ad} Z}$ with $Z \in \mathfrak{k}$. This conjugacy is a statement about the Lie algebra $\mathfrak{g}$ alone.

Next, let θ' be a second Cartan involution of $\mathfrak{g}$, with (-1)-eigenspace $\mathfrak{p}'$. By the uniqueness statement in § 20.1, $\theta' = \varphi\theta\varphi^{-1}$ for an inner automorphism φ. If $\theta X = -X$, then $\theta'(\varphi X) = \varphi\theta X = -\varphi X$. Symmetrically, $\theta = \varphi^{-1}\theta'\varphi$, so $\varphi^{-1}(\mathfrak{p}') \subseteq \mathfrak{p}$. Thus φ maps $\mathfrak{p}$ onto $\mathfrak{p}'$. Since φ is a Lie algebra automorphism, it maps maximal abelian subspaces of $\mathfrak{p}$ to maximal abelian subspaces of $\mathfrak{p}'$. Hence $\dim \mathfrak{a}$ is unchanged.

Theorem 20.8 (Cartan and Iwasawa decompositions; cited). *Let G be a connected semisimple Lie group with finite centre. Let θ be a Cartan involution of $\mathfrak{g}$, with $\mathfrak{g} = \mathfrak{k} \oplus \mathfrak{p}$, and let K be the connected subgroup with Lie algebra $\mathfrak{k}$. Choose $\mathfrak{a}$ and $\mathfrak{n}$ as above, and put $A = \exp \mathfrak{a}$ and $N = \exp \mathfrak{n}$. Then:*

(a) K is compact, and $K \times \mathfrak{p} \to G$, $(k, X) \mapsto k \exp X$, is a diffeomorphism;

(b) A and N are closed subgroups, and $K \times A \times N \to G$, $(k, a, n) \mapsto kan$, is a diffeomorphism.

For proofs see Knapp Beyond, Sections VI.3 and VI.4, or P. Etingof, *Lie Groups and Lie Algebras*, MIT 18.755 lecture notes, Lectures 15 and 18. The finite-centre hypothesis ensures that K is compact.

For $\mathrm{SL}(n, \mathbb{R})$ these decompositions are matrix factorisations. The form $G = KAN$ is the QR decomposition $g = k(an)$, where an is upper triangular with positive diagonal. Gram–Schmidt produces it (Exercise 20.2). The form $G = K \exp \mathfrak{p}$ writes $g = kP$ with $P = \exp X$ symmetric and positive definite. This is the polar decomposition of g, and $g^{T}g = P^{2} = \exp 2X$ determines P. In § 20.3 we prove both forms for $\mathrm{SL}(2, \mathbb{R})$, and we also obtain the singular value decomposition $G = KAK$.

A real form keeps the finite-dimensional complex modules of $\mathfrak{g}_{\mathbb{C}}$, by restriction and complex-linear extension. It does not keep their unitarity. For example, $\mathrm{SL}(2, \mathbb{R})$ acts on $V_m = \operatorname{Sym}^{m}\mathbb{C}^{2}$. For $m \geq 1$ and $t \neq 0$ the matrix $\operatorname{diag}(e^{t}, e^{-t})$ has the eigenvalue e^{mt} on V_m, and $|e^{mt}| \neq 1$. So no inner product makes V_m unitary for $\mathrm{SL}(2, \mathbb{R})$, whereas $\mathrm{SU}(2)$ has one by Lecture 5. To build unitary representations of $\mathrm{SL}(2, \mathbb{R})$ we use infinite-dimensional spaces, with K in Lecture 21 and KAN in Lecture 22.

In physics language, the adjoint action identifies $\mathfrak{sl}(2, \mathbb{R})$ with $\mathfrak{so}(2, 1)$. Then $\mathfrak{k}$ is the rotation generator and $\mathfrak{p}$ consists of the boosts, and $[\mathfrak{p}, \mathfrak{p}] \subset \mathfrak{k}$ says that the commutator of two boosts is a rotation (Lecture 24).

20.3 The rank-one model $\mathrm{SL}(2, \mathbb{R})$

Let $G = \mathrm{SL}(2, \mathbb{R})$ and $\theta(X) = -X^{T}$. By the description of $\mathfrak{sl}(n, \mathbb{R})$ in Proposition 20.5, $\mathfrak{k} = \mathbb{R}(E - F)$ and $\mathfrak{p} = \operatorname{span}(H, E + F)$. Take $\mathfrak{a} = \mathbb{R}H$. The restricted roots are $\pm\lambda_1$ with $\lambda_1(H) = 2$, and $\mathfrak{g}_{\lambda_1} = \mathbb{R}E$ and $\mathfrak{g}_{-\lambda_1} = \mathbb{R}F$. Hence the real rank is one. With $H_0 = H$ we get $\mathfrak{n} = \mathbb{R}E$. The subgroups K, A and N consist of the matrices

$$k_\varphi = \begin{pmatrix} \cos\varphi & -\sin\varphi \\ \sin\varphi & \cos\varphi \end{pmatrix}, \qquad a_t = \begin{pmatrix} e^{t} & 0 \\ 0 & e^{-t} \end{pmatrix}, \qquad n_x = \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix},$$

with $K = \mathrm{SO}(2) = \{k_\varphi\}$, $A = \{a_t\}$ and $N = \{n_x\}$. We now prove the conclusions of the Cartan and Iwasawa theorem, Theorem 20.8, directly for G, without using that theorem.

Proposition 20.9 (Iwasawa decomposition of $\mathrm{SL}(2,\mathbb{R})$)**.** *The map $K\times A\times N \to G$, $(k,a,n)\mapsto kan$, is a diffeomorphism. Explicitly, for*

$$g = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \in G, \qquad r = \sqrt{\alpha^2+\gamma^2},$$

one has $g = k_\varphi a_t n_x$ with

$$\cos\varphi = \frac{\alpha}{r}, \qquad \sin\varphi = \frac{\gamma}{r}, \qquad e^t = r, \qquad x = \frac{\alpha\beta+\gamma\delta}{r^2}.$$

Proof. The first column of g is nonzero, so $r>0$. Let $k = k_\varphi$ with φ as stated. This is Gram–Schmidt applied to the columns of g. Using $\alpha\delta-\beta\gamma = 1$,

$$k^{-1}g = \frac{1}{r}\begin{pmatrix} \alpha & \gamma \\ -\gamma & \alpha \end{pmatrix}\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} = \begin{pmatrix} r & \dfrac{\alpha\beta+\gamma\delta}{r} \\ 0 & \dfrac{1}{r} \end{pmatrix}.$$

On the other hand,

$$a_t n_x = \begin{pmatrix} e^t & e^t x \\ 0 & e^{-t} \end{pmatrix}.$$

These agree for $e^t = r$ and $x = (\alpha\beta+\gamma\delta)/r^2$. This proves existence.

For uniqueness, suppose $kan = k'a'n'$. Then $k'^{-1}k = a'n'(an)^{-1}$ is orthogonal and upper triangular with positive diagonal. Its first column is $(p,0)^{T}$ with $p>0$ and unit length, so $p=1$. The second column is a unit vector orthogonal to $(1,0)^{T}$ with positive second entry, so it is $(0,1)^{T}$. Hence $k=k'$ and $an = a'n'$. Comparing diagonals gives $a=a'$, and then $n=n'$.

The multiplication map is smooth. Its inverse is given by the formulas above, which are smooth in the entries of g because $r>0$. ∎

In particular G is connected and diffeomorphic to a circle times $\mathbb{R}^2$. So $\mathrm{SL}(2,\mathbb{R})$ satisfies the hypotheses of the Cartan and Iwasawa theorem, Theorem 20.8, with centre $\{\pm I\}$.

For the Cartan decomposition we need the exponential on $\mathfrak{p}$. Define the entire functions

$$C(u) = \sum_{k\ge 0}\frac{u^k}{(2k)!}, \qquad S(u) = \sum_{k\ge 0}\frac{u^k}{(2k+1)!}.$$

For $u = s^2 \ge 0$ they are $\cosh s$ and $\sinh s/s$, with the latter interpreted as 1 at $s=0$. We use $u=s^2$ rather than $s=\sqrt{-\det Y}$, because s is not smooth at $Y=0$. Let $\mathcal{P}$ be the set of symmetric positive definite 2×2 matrices of determinant 1.

Lemma 20.10. *For $Y\in\mathfrak{p}$, set $u = -\det Y \ge 0$. Then $Y^2 = uI$ and*

$$\exp Y = C(u)I + S(u)Y.$$

The map $\exp\colon \mathfrak{p} \to \mathcal{P}$ *is a bijection. Its inverse is*

$$L(Q) = \frac{Q - cI}{\mathcal{S}(C^{-1}(c))}, \qquad c = \tfrac{1}{2}\operatorname{tr} Q,$$

which is smooth on a neighbourhood of $\mathcal{P}$ *among symmetric matrices.*

Proof. Write $Y = \begin{pmatrix} p & q \\ q & -p \end{pmatrix}$. Then $Y^2 = (p^2 + q^2)I = uI$. Splitting the exponential series into even and odd powers gives the formula. The matrix $\exp Y = (\exp \frac{Y}{2})^2$ is the square of an invertible symmetric matrix. Hence $\exp Y$ is positive definite. Moreover, $\det \exp Y = e^{\operatorname{tr} Y} = 1$.

Every coefficient of the series for C' is positive, so $C' > 0$ on $(0, \infty)$. Also, $C'(0) = \frac{1}{2}$. So $C' > 0$ on some interval $(-\varepsilon, \infty)$. Thus C is a strictly increasing bijection $[0, \infty) \to [1, \infty)$, with smooth inverse near $[1, \infty)$. Also $\mathcal{S} > 0$ near $[0, \infty)$.

For injectivity, let $Q = \exp Y$. Then $\operatorname{tr} Q = 2C(u)$, which determines u because C is strictly increasing on $[0, \infty)$. The formula for $\exp Y$ then gives $Y = (Q - C(u)I)/\mathcal{S}(u) = L(Q)$.

For surjectivity, let $Q \in \mathcal{P}$ have eigenvalues $\mu, \mu^{-1} > 0$. Then $c = \frac{1}{2}(\mu + \mu^{-1}) \geq 1$. Put $u = C^{-1}(c)$ and $Y = L(Q)$. This Y is symmetric and traceless. Cayley–Hamilton gives $Q^2 = 2cQ - I$, so $(Q - cI)^2 = (c^2 - 1)I$. Since $u\,\mathcal{S}(u)^2 = C(u)^2 - 1 = c^2 - 1$, we get $Y^2 = uI$. Therefore

$$\exp Y = C(u)I + \mathcal{S}(u)Y = cI + (Q - cI) = Q. \qquad \blacksquare$$

Proposition 20.11 (Cartan decomposition of $\mathrm{SL}(2, \mathbb{R})$)**.** *The map* $K \times \mathfrak{p} \to G$, $(k, X) \mapsto k \exp X$, *is a diffeomorphism. Its inverse is*

$$g \mapsto \big(g \exp(-X),\, X\big), \qquad X = \tfrac{1}{2} L(g^T g).$$

Proof. Suppose first that $g = k \exp X$ with $k \in K$. Since X is symmetric, so is $\exp X$. Also k is orthogonal. Hence

$$g^T g = \exp(X) k^T k \exp(X) = \exp(2X).$$

By the exponential lemma, Lemma 20.10, $2X = L(g^T g)$, so X and then k are forced. Conversely, given g, the matrix $g^T g$ lies in $\mathcal{P}$. Define X and $k = g \exp(-X)$ as stated. Then

$$k^T k = \exp(-X) \exp(2X) \exp(-X) = I, \qquad \det k = 1,$$

so $k \in K$. The map and its inverse are smooth by the exponential lemma, Lemma 20.10. $\blacksquare$

Corollary 20.12 ($G = KAK$)**.** *Every* $g \in G$ *can be written as* $g = k_1 a_t k_2$ *with* $k_1, k_2 \in K$ *and* $t \geq 0$*, and* t *is unique. The number* e^{2t} *is the larger eigenvalue of* $g^T g$*. So* e^t *is the larger singular value of* g*.*

Proof. Write $g = k \exp X$. The symmetric traceless matrix X has eigenvalues $\pm t$ with $t \geq 0$. Take an orthonormal eigenbasis, ordered $(+t, -t)$, and negate one vector if needed. This gives $k' \in K$ with $X = k'(tH)k'^{-1}$. Then $\exp X = k' a_t k'^{-1}$ and $g = (kk') a_t k'^{-1}$. If $g = k_1 a_t k_2$, then $g^T g = k_2^{-1} a_{2t} k_2$ has eigenvalues $e^{\pm 2t}$, which fixes t. $\blacksquare$

Example 20.13. We compute the Iwasawa, Cartan and KAK data of the lower triangular unipotent matrix

$$g = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}.$$

The Cartan data involve the golden ratio $\tau = \frac{1+\sqrt{5}}{2}$. Recall $\tau + \tau^{-1} = \sqrt{5}$ and $\tau - \tau^{-1} = 1$.

Iwasawa. Here $r = \sqrt{2}$, so $k = k_{\pi/4}$, $e^t = \sqrt{2}$ and $x = \tfrac{1}{2}$. Check:

$$\frac{1}{\sqrt{2}}\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}\begin{pmatrix} \sqrt{2} & \frac{\sqrt{2}}{2} \\ 0 & \frac{1}{\sqrt{2}} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}.$$

Cartan. Here

$$Q = g^{T} g = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}.$$

The half-trace of Q is $c = \tfrac{3}{2}$. The equation $C(u) = \cosh\sqrt{u} = \tfrac{3}{2}$ gives $\sqrt{u} = 2\log\tau$, since $\tau^2 + \tau^{-2} = 3$. Also $\sinh\sqrt{u} = \frac{\sqrt{5}}{2}$. Hence

$$X = \frac{1}{2}L(Q) = \frac{2\log\tau}{\sqrt{5}}\begin{pmatrix} \frac{1}{2} & 1 \\ 1 & -\frac{1}{2} \end{pmatrix} = \frac{\log\tau}{\sqrt{5}}\begin{pmatrix} 1 & 2 \\ 2 & -1 \end{pmatrix}.$$

Now $-\det X = (\log\tau)^2$, so the exponential lemma, Lemma 20.10, gives

$$\exp X = \frac{\sqrt{5}}{2} I + \frac{1}{2\sqrt{5}}\begin{pmatrix} 1 & 2 \\ 2 & -1 \end{pmatrix} = \frac{1}{\sqrt{5}}\begin{pmatrix} 3 & 1 \\ 1 & 2 \end{pmatrix}.$$

Then

$$k = g\exp(-X) = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}\frac{1}{\sqrt{5}}\begin{pmatrix} 2 & -1 \\ -1 & 3 \end{pmatrix} = \frac{1}{\sqrt{5}}\begin{pmatrix} 2 & -1 \\ 1 & 2 \end{pmatrix} \in \mathrm{SO}(2).$$

The eigenvalues of X are $\pm\log\tau$. Hence, as in the proof of the KAK corollary, Corollary 20.12, the KAK parameter of g is $t = \log\tau$.

20.4 The upper half-plane

Let $\mathbb{H} = \{z \in \mathbb{C} : \operatorname{Im} z > 0\}$. For g with entries $\alpha, \beta, \gamma, \delta$ as in the Iwasawa decomposition, Proposition 20.9, put

$$g \cdot z = \frac{\alpha z + \beta}{\gamma z + \delta}.$$

Lemma 20.14. *For $z \in \mathbb{H}$ one has $\gamma z + \delta \neq 0$ and*

$$\operatorname{Im}(g \cdot z) = \frac{\operatorname{Im} z}{|\gamma z + \delta|^2}.$$

This defines an action of G on $\mathbb{H}$. The stabiliser of i is K, and $-I$ acts trivially.

Proof. If $\gamma \neq 0$, the root of $\gamma z + \delta$ is real. If $\gamma = 0$, then $\delta \neq 0$. Next,

$$g \cdot z = \frac{(\alpha z + \beta)(\gamma \bar{z} + \delta)}{|\gamma z + \delta|^2},$$

and the numerator has imaginary part $(\alpha\delta - \beta\gamma)\operatorname{Im} z = \operatorname{Im} z$.

For the action law, write $[v] = v_1/v_2$ for $v \in \mathbb{C}^2$ with $v_2 \neq 0$. Then $g \cdot z = [g(z,1)^T]$, and $[\lambda v] = [v]$ for $\lambda \neq 0$. If h has bottom row (γ', δ'), then $h(z,1)^T = (\gamma' z + \delta')(h \cdot z, 1)^T$. Hence

$$(gh) \cdot z = [gh(z,1)^T] = [g(h \cdot z, 1)^T] = g \cdot (h \cdot z).$$

If $g \cdot i = i$, then $\alpha i + \beta = -\gamma + \delta i$. So $\alpha = \delta$ and $\beta = -\gamma$, with $\alpha^2 + \gamma^2 = 1$. That is, $g = k_\varphi$. Conversely, $k_\varphi \cdot i = ie^{i\varphi}/e^{i\varphi} = i$. Finally, $(-I) \cdot z = z$. ∎

Inverting $g^{-1} = kan$ gives $g = n^{-1}a^{-1}k^{-1}$. So the Iwasawa decomposition, Proposition 20.9, also yields a unique decomposition $G = NAK$.

Proposition 20.15. *Write $g = n_x a_t k$. Then*

$$g \cdot i = x + ie^{2t}, \qquad x = \frac{\alpha\gamma + \beta\delta}{\gamma^2 + \delta^2}, \qquad e^{2t} = \frac{1}{\gamma^2 + \delta^2}.$$

The map $n_x a_t \mapsto n_x a_t \cdot i$ is a diffeomorphism $NA \to \mathbb{H}$. The map $gK \mapsto g \cdot i$ is a bijection $G/K \to \mathbb{H}$.

These numbers x and t are not those of the Iwasawa decomposition, Proposition 20.9, of g. From $g = n_x a_t k$ we get $g^{-1} = k^{-1}a_{-t}n_{-x}$. So they are the negatives of the parameters of g^{-1} in $G = KAN$.

Proof. We have $a_t \cdot z = e^{2t}z$ and $n_x \cdot z = z + x$. Since $k \cdot i = i$, this gives $n_x a_t k \cdot i = x + ie^{2t}$. The stated formulas for x and e^{2t} follow by comparing real and imaginary parts in the direct computation

$$g \cdot i = \frac{(\alpha i + \beta)(\delta - \gamma i)}{\gamma^2 + \delta^2} = \frac{\alpha\gamma + \beta\delta + i}{\gamma^2 + \delta^2}.$$

The map $(x,t) \mapsto x + ie^{2t}$ has smooth inverse $z \mapsto (\operatorname{Re} z, \frac{1}{2}\log \operatorname{Im} z)$. Finally, $g \cdot i = g' \cdot i$ exactly when $g^{-1}g' \in K$, by the stabiliser statement of Lemma 20.14. ∎

The action of G on $\mathbb{H}$ is smooth and transitive. So, by the homogeneous-space theorem of Lecture 3, the bijection $G/K \to \mathbb{H}$ is a diffeomorphism.

So Iwasawa coordinates are Cartesian coordinates (x, y) with $y = e^{2t}$. Here N translates horizontally and A dilates from 0. The subgroup NA acts by the affine maps $z \mapsto e^{2t}z + x$; Exercise 20.6 identifies it with the affine group of Lecture 16. Under this identification the left Haar measure $dy\,db/y^2$ of Lecture 16 becomes the measure $dx\,dy/y^2$ on $\mathbb{H}$.

The KAK form describes $\mathbb{H}$ by K-orbits. If $g \cdot i = z$ and $g = k_1 a_t k_2$, then $z = k_1 \cdot (ie^{2t})$. The number $t \geq 0$ depends only on z, because replacing g by gk does not change the eigenvalues of $g^T g$. So $\mathbb{H}$ is the disjoint union of the K-orbits of the points ie^{2t}, $t \geq 0$. Exercise 20.5 shows that these orbits are Euclidean circles.

Figure 20.1 draws both descriptions. It marks the point $g \cdot i = \frac{1}{2}(1 + i)$ for the matrix g of Example 20.13. This point has Iwasawa coordinates $x = \frac{1}{2}$ and $e^{2t} = \frac{1}{2}$. Its KAK parameter

is $t = \log \tau$, so it lies on the K-orbit of $i\tau^2$. The derivative of $z \mapsto k_\varphi \cdot z$ at i is $e^{-2i\varphi}$, so k_φ with $\varphi > 0$ turns clockwise around i.

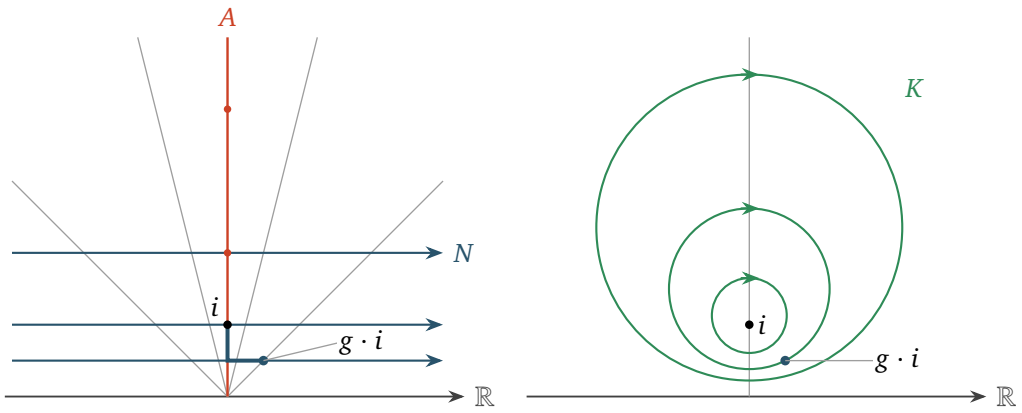


Figure 20.1: Coordinates on $\mathbb{H}$, with the point $g \cdot i$ of the lower triangular example. The first panel shows N-orbits (horizontal lines), A-orbits (rays from 0, with the imaginary axis through i) and a heavy path from i along A and then along N. The second panel shows K-orbits (circles, and the point i itself), with arrows for k_φ, $\varphi > 0$.

20.5 Self-checks

1. What are the real ranks of $\mathfrak{sl}(n, \mathbb{R})$ and $\mathfrak{su}(n)$? What do A and N become for $\mathrm{SU}(n)$?
   Answer. The real rank of $\mathfrak{sl}(n, \mathbb{R})$ is $n - 1$, and the real rank of $\mathfrak{su}(n)$ is 0. For $\mathrm{SU}(n)$, $\mathfrak{p} = 0$, so $A = N = \{I\}$ and the Iwasawa decomposition reads $G = K$.

2. Find the Iwasawa decomposition of $g = \begin{pmatrix} 0 & 1 \\ -1 & 2 \end{pmatrix}$.
   Answer. Here $r = 1$, $\cos\varphi = 0$ and $\sin\varphi = -1$, so $k = k_{-\pi/2}$ and $a = I$. Also $x = (0 \cdot 1 + (-1) \cdot 2)/1 = -2$. Indeed
   $$\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} 1 & -2 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -1 & 2 \end{pmatrix}.$$

3. Where does $-I$ sit in $G = KAN$, and what does it do on $\mathbb{H}$?
   Answer. In $G = KAN$, the element $-I$ has factors $k = k_\pi$ and $a = n = I$. The element $-I$ acts trivially on $\mathbb{H}$. Hence the action factors through $G/\{\pm I\}$.

Summary

A Cartan involution splits $\mathfrak{g} = \mathfrak{k} \oplus \mathfrak{p}$. The compact form $\mathfrak{k} \oplus i\mathfrak{p}$ of $\mathfrak{g}_{\mathbb{C}}$ has real rank zero. The split form $\mathfrak{sl}(n, \mathbb{R})$ has real rank $n - 1$. For connected semisimple groups with finite centre, the cited Cartan and Iwasawa theorem gives $G = K \exp \mathfrak{p}$ and $G = KAN$. For $\mathrm{SL}(2, \mathbb{R})$ we proved both decompositions, with explicit inverses. On $\mathbb{H}$, identified with G/K, the subgroup NA gives Cartesian coordinates. The KAK form describes $\mathbb{H}$ as the disjoint union of the K-orbits of the points ie^{2t} with $t \geq 0$.

Reading

- Knapp Beyond, Chapter VI, Sections 1–4: compact real forms, the Cartan decomposition on the Lie algebra and group levels, and the Iwasawa decomposition.
- P. Etingof, *Lie Groups and Lie Algebras*, MIT 18.755 lecture notes, Lectures 15 and 18: a second route through real forms and these decompositions.
- Lecture 22 uses $G = KAN$ to construct the principal series.